# The Sequential Monoidal Structure on Containers

Extracted from a machine-checked development

**Abstract**

The category **Cont** of containers and container morphisms carries a monoidal structure whose tensor is the sequential (composition) operator $\triangleleft$ and whose unit is the identity container $I$. We record the bifunctoriality of $\triangleleft$ on morphisms, the naturality of the associator and unitors, and the two coherence laws—the pentagon and the triangle. All equalities are transport-free: position fibres line up definitionally, so every law holds by direct computation. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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## 1 Introduction

Containers, in the sense of Abbott, Altenkirch, and Ghani, form a category **Cont** whose objects are pairs $(S, P)$ of a shape type and a position family, and whose morphisms are shape maps with contravariant position maps. Composition of the associated polynomial functors lifts to a tensor product on containers: the *sequential operator* $G \triangleleft F$ puts an $F$-structure at every position of a $G$-structure. Its unit is the identity container $I$ (one shape, one position).

The structural isomorphisms—associator and left/right unitors—are standard and transport-free. What remains to certify $(\mathbf{Cont}, \triangleleft, I)$ as a monoidal category is bifunctoriality of $\triangleleft$ on morphisms and the two Mac Lane coherence laws. That is the content of this note.

**What this note claims.** In ordinary mathematical language: $\triangleleft$ is a bifunctor; the associator and unitors are natural; the pentagon and triangle commute. On shapes the associator is currying; on positions it is $\Sigma$-reassociation. Both legs of the pentagon flatten the fourfold composite $((C_1 \triangleleft C_2) \triangleleft C_3) \triangleleft C_4$ to the right-nested normal form $C_1 \triangleleft (C_2 \triangleleft (C_3 \triangleleft C_4))$, and on positions every map is the identity on the underlying flat tuple.

We work throughout in a single universe of types. Proofs are indicated by the phrase "by lean."

## 2 Background

**Definition 2.1** (Container)**.** A **container** $C$ consists of a type $\mathrm{Shape}(C)$ of shapes and, for each shape $s$, a type $\mathrm{Pos}(C)(s)$ of positions.

**Definition 2.2** (Container morphism)**.** A **morphism** $\mu : C \to D$ consists of a shape map $u : \mathrm{Shape}(C) \to \mathrm{Shape}(D)$ and, for each $s$, a position map $\mathrm{Pos}(D)(u(s)) \to \mathrm{Pos}(C)(s)$. Composition and identities are the evident ones; we write **Cont** for the resulting category.

**Definition 2.3** (Sequential operator)**.** The **sequential composite** $G \triangleleft F$ of containers $G$ and $F$ has

$$\mathrm{Shape}(G \triangleleft F) \;=\; \sum_{t \,:\, \mathrm{Shape}(G)} \big(\mathrm{Pos}(G)(t) \to \mathrm{Shape}(F)\big), \tag{1}$$

$$\mathrm{Pos}(G \triangleleft F)\langle t,\, f\rangle \;=\; \sum_{p \,:\, \mathrm{Pos}(G)(t)} \mathrm{Pos}(F)\big(f(p)\big). \tag{2}$$

An element of the extension $\mathrm{Ext}\,(G \triangleleft F)\,X$ is a $G$-structure of $F$-structures of $X$-labels—equivalently, $\mathrm{Ext}\,G\big(\mathrm{Ext}\,F\,X\big)$.

**Definition 2.4** (Unit container)**.** The **identity container** $I$ has a single shape and a single position at that shape. It is the unit for $\triangleleft$.

**Remark 2.5** (Structural isomorphisms)**.** The associator

$$\alpha_{X,Y,Z} \;:\; (X \triangleleft Y) \triangleleft Z \;\xrightarrow{\sim}\; X \triangleleft (Y \triangleleft Z)$$

acts by currying on shapes and by $\Sigma$-reassociation on positions. The left unitor $\lambda_X : I \triangleleft X \xrightarrow{\sim} X$ and right unitor $\rho_X : X \triangleleft I \xrightarrow{\sim} X$ are the evident collapses. All three are isomorphisms of containers (transport-free), and we take them as given from the sequential development.

## 3 Bifunctoriality

**Definition 3.1** (Action on morphisms)**.** Given morphisms $\mu : G \to G'$ and $\nu : F \to F'$, the **sequential action**

$$\mu \triangleleft_2 \nu \;:\; G \triangleleft F \;\longrightarrow\; G' \triangleleft F'$$

is the container morphism with

$$\text{on shapes:}\quad \langle t,\, f\rangle \;\longmapsto\; \langle \mu(t),\, \nu \circ f \circ \mu^\sharp_t\rangle, \tag{3}$$

$$\text{on positions:}\quad \langle p,\, q\rangle \;\longmapsto\; \langle \mu^\sharp_t(p),\, \nu^\sharp_{f(\mu^\sharp_t(p))}(q)\rangle, \tag{4}$$

where $\mu^\sharp$ and $\nu^\sharp$ denote the (contravariant) position components of $\mu$ and $\nu$. The position fibres of source and target agree definitionally, so no transport appears.

**Definition 3.2** (Whiskering)**.**

1. **Left whiskering** $G \triangleleft \nu : G \triangleleft F \to G \triangleleft F'$ is $\mathrm{id}_G \triangleleft_2 \nu$.
2. **Right whiskering** $\mu \triangleleft F : G \triangleleft F \to G' \triangleleft F$ is $\mu \triangleleft_2 \mathrm{id}_F$.

**Proposition 3.3** (Functor laws)**.** *For all containers and morphisms of the appropriate types:*

1. $\mathrm{id}_G \triangleleft_2 \mathrm{id}_F = \mathrm{id}_{G \triangleleft F}$*;*
2. $(\mu\,;\mu') \triangleleft_2 (\nu\,;\nu') = (\mu \triangleleft_2 \nu)\,;(\mu' \triangleleft_2 \nu')$ *(interchange).*

*Proof.* By lean. □

Thus $\triangleleft$ is a bifunctor $\mathbf{Cont} \times \mathbf{Cont} \to \mathbf{Cont}$.

## 4 Naturality of the structural isomorphisms

**Proposition 4.1** (Naturality of the associator)**.** *For morphisms $f_i : X_i \to Y_i$ $(i = 1, 2, 3)$,*

$$\big((f_1 \lhd_2 f_2) \lhd_2 f_3\big) \,;\, \alpha_{Y_1,Y_2,Y_3} \;=\; \alpha_{X_1,X_2,X_3} \,;\, \big(f_1 \lhd_2 (f_2 \lhd_2 f_3)\big). \tag{5}$$

*Proof.* By lean. □

**Proposition 4.2** (Naturality of the unitors)**.** *For every morphism $f : X \to Y$,*

$$(I \lhd f) \,;\, \lambda_Y \;=\; \lambda_X \,;\, f, \tag{6}$$

$$(f \lhd I) \,;\, \rho_Y \;=\; \rho_X \,;\, f. \tag{7}$$

*Proof.* By lean. □

**Remark 4.3.** Each square holds because both legs are the same combination of currying/$\Sigma$-reassociation with the morphism data threaded identically.

## 5 Coherence: pentagon and triangle

**Theorem 5.1** (Pentagon identity)**.** *For all containers $C_1, C_2, C_3, C_4$, the two ways of reassociating $((C_1 \lhd C_2) \lhd C_3) \lhd C_4$ to $C_1 \lhd (C_2 \lhd (C_3 \lhd C_4))$ coincide:*

$$\begin{aligned} &\big((\alpha_{C_1,C_2,C_3} \lhd C_4) \,;\, \alpha_{C_1,\, C_2 \lhd C_3,\, C_4}\big) \,;\, (C_1 \lhd \alpha_{C_2,C_3,C_4}) \\ &\quad= \; \alpha_{C_1 \lhd C_2,\, C_3,\, C_4} \,;\, \alpha_{C_1,\, C_2,\, C_3 \lhd C_4}. \end{aligned} \tag{8}$$

*Proof.* By lean. □

**Remark 5.2.** Both legs flatten the left-nested fourfold composite to the right-nested normal form. On positions, every map is the identity on the underlying flat tuple of positions, so the two paths agree on the nose.

**Theorem 5.3** (Triangle identity)**.** *For all containers $C_1, C_2$,*

$$\alpha_{C_1,\, I,\, C_2} \,;\, (C_1 \lhd \lambda_{C_2}) \;=\; \rho_{C_1} \lhd C_2, \tag{9}$$

*as morphisms $(C_1 \lhd I) \lhd C_2 \to C_1 \lhd C_2$.*

*Proof.* By lean. □

## 6 The monoidal category

**Definition 6.1** (Isomorphism)**.** An **isomorphism** $X \cong Y$ in a category is a morphism with a two-sided inverse.

**Definition 6.2** (Monoidal category)**.** A **monoidal category** consists of a category $\mathcal{C}$ together with

1. a tensor product $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ (an object map and a morphism map satisfying the two functor laws),
2. a unit object $I$,
3. natural isomorphisms $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \xrightarrow{\sim} X \otimes (Y \otimes Z)$, $\lambda_X : I \otimes X \xrightarrow{\sim} X$, $\rho_X : X \otimes I \xrightarrow{\sim} X$,

4. the pentagon and triangle coherence laws.

**Theorem 6.3** ((**Cont**, $\triangleleft$, $I$) is monoidal)**.** *The category of containers, equipped with the sequential operator as tensor, the identity container as unit, and the structural isomorphisms recalled above, is a monoidal category. Bifunctoriality is Proposition 3.3; naturality is Propositions 4.1 and 4.2; coherence is Theorems 5.1 and 5.3.*

*Proof.* By lean. □

**Remark 6.4.** Every law holds by direct computation: the development is transport-free and axiom-clean. In particular there is no appeal to Mac Lane's coherence theorem as a black box; the pentagon and triangle are checked on the nose for this concrete tensor.

# 7 Discussion

**What has been established.** The sequential operator makes **Cont** into a monoidal category. The structure is the spine of the broader development: comonoids for (**Cont**, $\triangleleft$, $I$) are precisely directed containers, so the monoidal data certified here underwrite that correspondence.

**Relation to other monoidal structures.** The category **Cont** carries at least four monoidal structures of interest (coproduct, product, Dirichlet tensor, and sequential composition). Only one can be the canonical monoidal instance without colliding in a formal system; the sequential structure is preferred here because it is the composition of polynomial functors. The other three are treated in companion notes.

**Machine checking.** Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.


# Acknowledgements

The mathematical content follows the theory of containers of Abbott, Altenkirch, and Ghani, and the monoidal structure of sequential composition of polynomial functors. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.


# References


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